

Some review questions - 1st week

Q1. *Setting $u = (-1, 3)$ and $v = (0, 1)$, compute $u + v$ and represent u , v and $u + v$ on the plane. Give the general definition of the sum of two given vectors $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and $y = (y_1, \dots, y_p) \in \mathbb{R}^p$.*

Q2. *Setting $u = (-1, 2)$, compute the norm of u . Give the general definition and the geometric interpretation of norm for given $x = (x_1, \dots, x_p) \in \mathbb{R}^p$.*

Q3. *Translate each statement below into one of the mathematical expressions: $P \Rightarrow Q$, $Q \Rightarrow P$, or $P \Leftrightarrow Q$, where P is “ $x^2 = 9$ ” and Q is “ $x = 3$ ”. Judge whether each statement is true or false.*

- *if $x = 3$ then $x^2 = 9$*
- *$x = 3$ if $x^2 = 9$*
- *$x^2 = 9$ if and only if $x = 3$*
- *$x^2 = 9$ only if $x = 3$*

Q4. *Given a vector $u \in \mathbb{R}^p$, prove that $\|u\| = 0$ if and only if $u = 0$.*

Q5. *Given a vector u with size p and a scalar c , give a simple expression of $\|cu\|$ and prove your result.*

Q6. *Give the definition and some examples of unit vectors. Prove that for all nonzero vectors v , $\frac{v}{\|v\|}$ is a unit vector.*

Q7. Setting $u = (-1, 3, 1)$ and $v = (0, 1, -3)$, compute $\langle u, v \rangle$. Give the general definition of the inner product of any two vectors $x = (x_1, \dots, x_p)$ and $y = (y_1, \dots, y_p)$.

Q8. Give the relationship between the inner product and the norm.

Q8. Explain how the final score of MAT2041 is related to inner product.